

Coordinate Proofs and Bisector Theorems

Coordinate Proofs

① Show that the triangle below is isosceles.

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D_{LQ} = \sqrt{(7-0)^2 + (8-7)^2}$$

$$D_{LQ} = \sqrt{49 + 1}$$

$$D_{LQ} = \sqrt{50}$$

$$D_{LQ} = 5\sqrt{5}$$

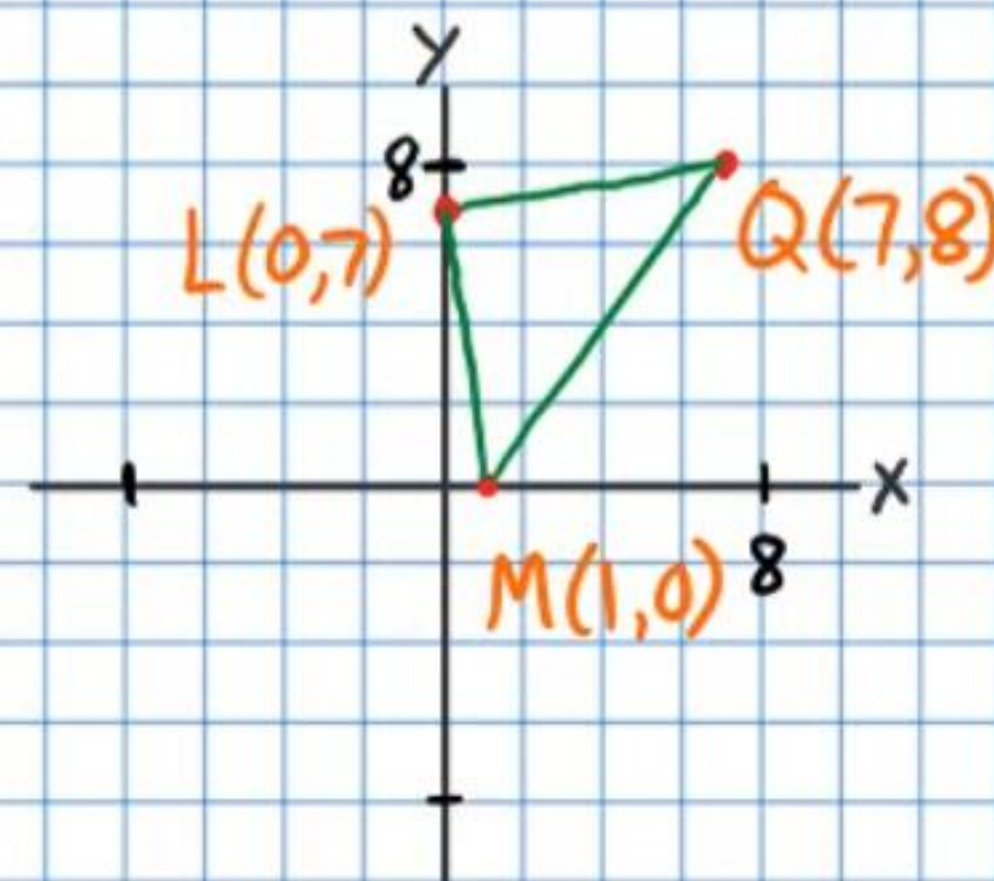
$$D_{LM} = \sqrt{(1-0)^2 + (0-7)^2}$$

$$D_{LM} = \sqrt{1 + 49}$$

$$D_{LM} = \sqrt{50}$$

$$D_{LM} = 5\sqrt{5}$$

Therefore, $\triangle LQM$ is isosceles.



② Show that the triangle below is isosceles.

$$D_{TR} = \sqrt{(6-4)^2 + (2-8)^2}$$

$$D_{TR} = \sqrt{4 + 36}$$

$$D_{TR} = \sqrt{40}$$

$$D_{TR} = 2\sqrt{10}$$

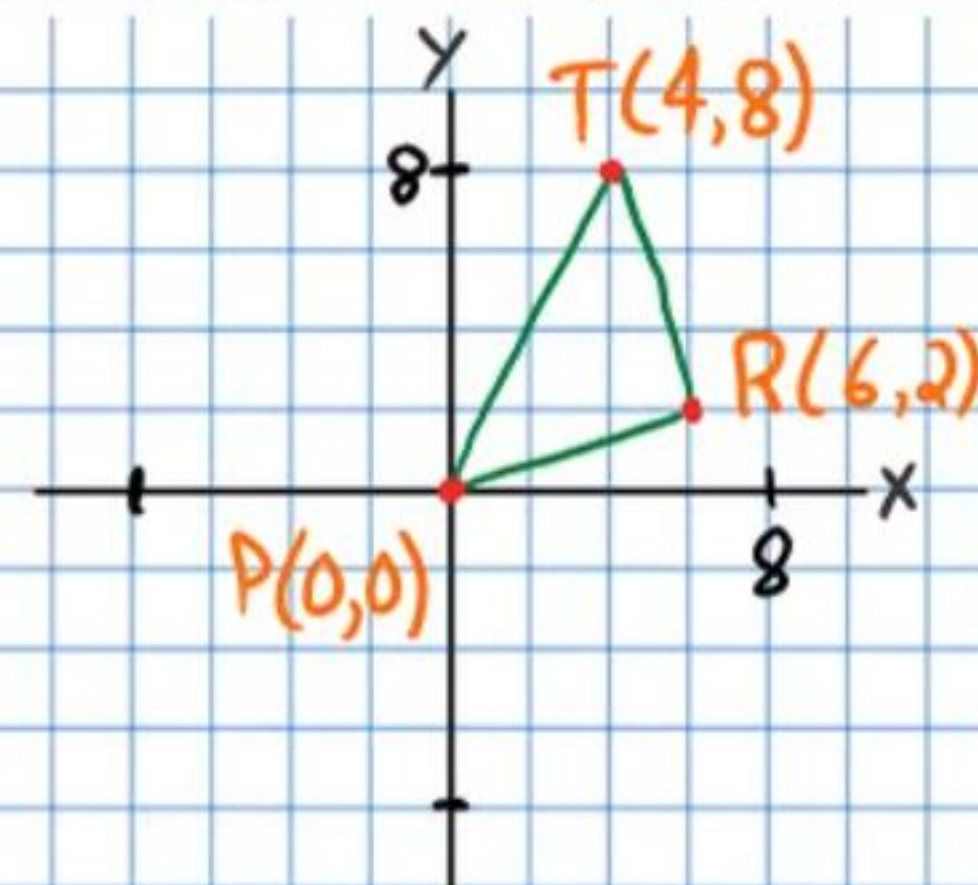
$$D_{PR} = \sqrt{(6-0)^2 + (2-0)^2}$$

$$D_{PR} = \sqrt{36 + 4}$$

$$D_{PR} = \sqrt{40}$$

$$D_{PR} = 2\sqrt{10}$$

Therefore, $\triangle TRP$ is isosceles.



③ Prove that the diagonals of the quadrilateral below bisect each other.

$$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

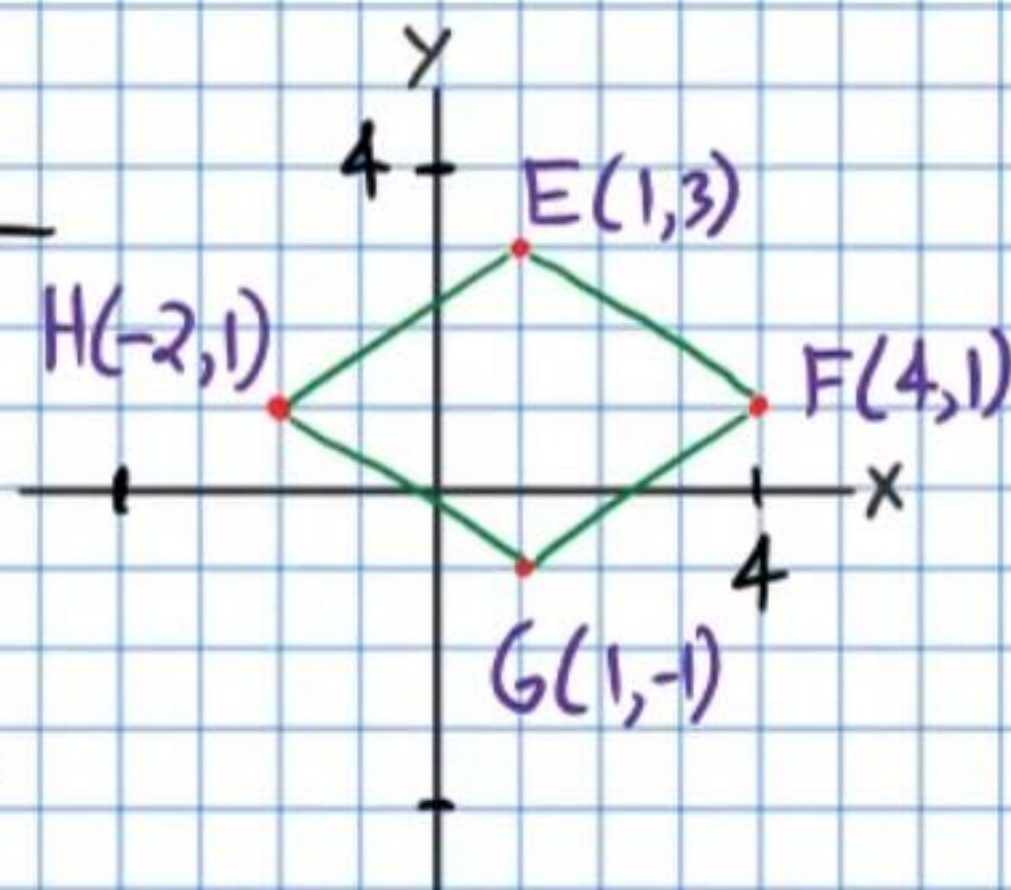
$$M_{HF}\left(\frac{-2+4}{2}, \frac{1+1}{2}\right)$$

$$M_{HF}(1, 1)$$

$$M_{EG}\left(\frac{1+1}{2}, \frac{3+(-1)}{2}\right)$$

$$M_{EG}(1, 1)$$

Therefore, the diagonals bisect each other.



④ Prove that the diagonals of the quadrilateral below bisect each other.

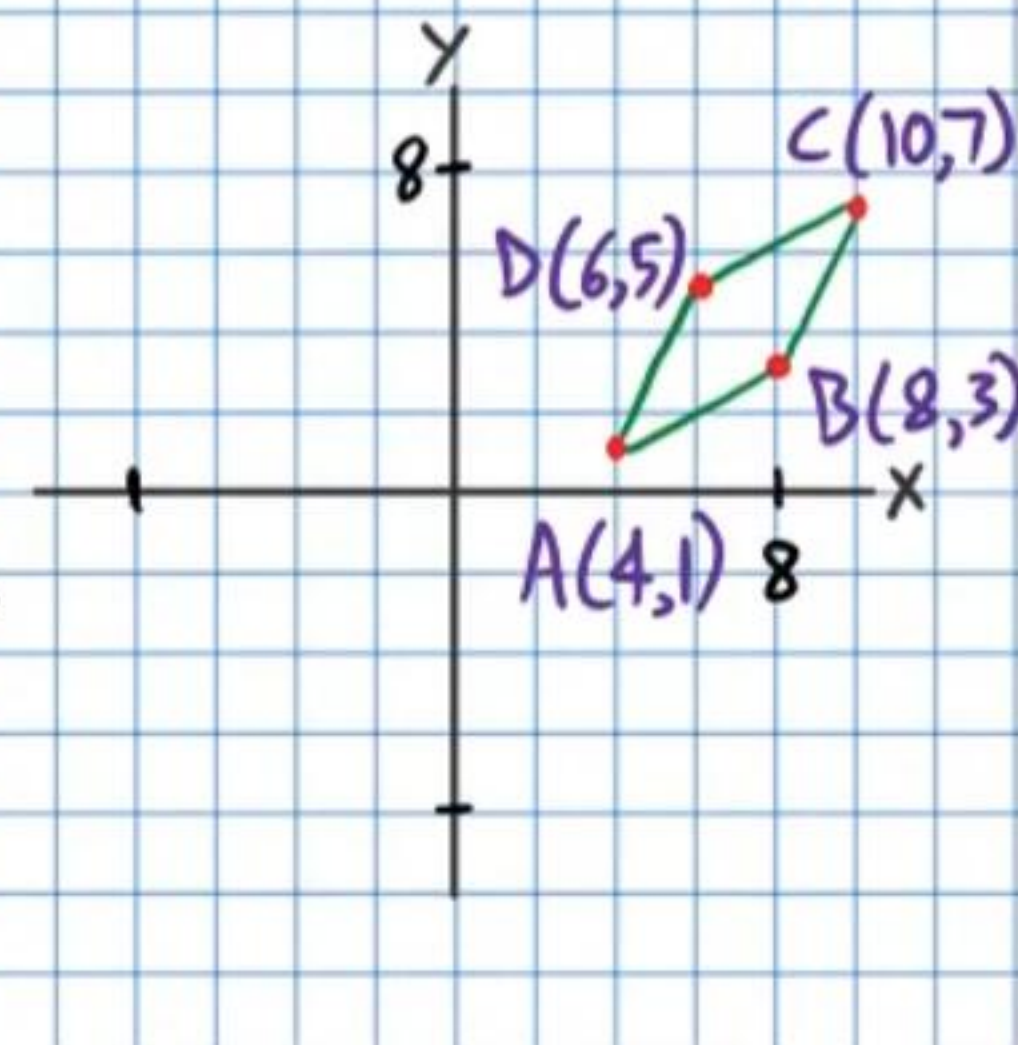
$$M_{DB}\left(\frac{6+8}{2}, \frac{5+3}{2}\right)$$

$$M_{DB}(7, 4)$$

$$M_{AC}\left(\frac{4+10}{2}, \frac{1+7}{2}\right)$$

$$M_{AC}(7, 4)$$

Therefore, the diagonals bisect each other.



Review

Slopes of Parallel Line Segments

If the slopes of line segments are equal, then the line segments are parallel.

Slope

The slope of a line segment containing points (x_1, y_1) and (x_2, y_2) is given by the following formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Slopes of Perpendicular Line Segments

If the slopes of two line segments are opposite reciprocals, then the measure of the angle between the line segments is 90° .

Parallelogram: A quadrilateral with two pairs of parallel sides.

⑤ Show that the quadrilateral below is a parallelogram.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

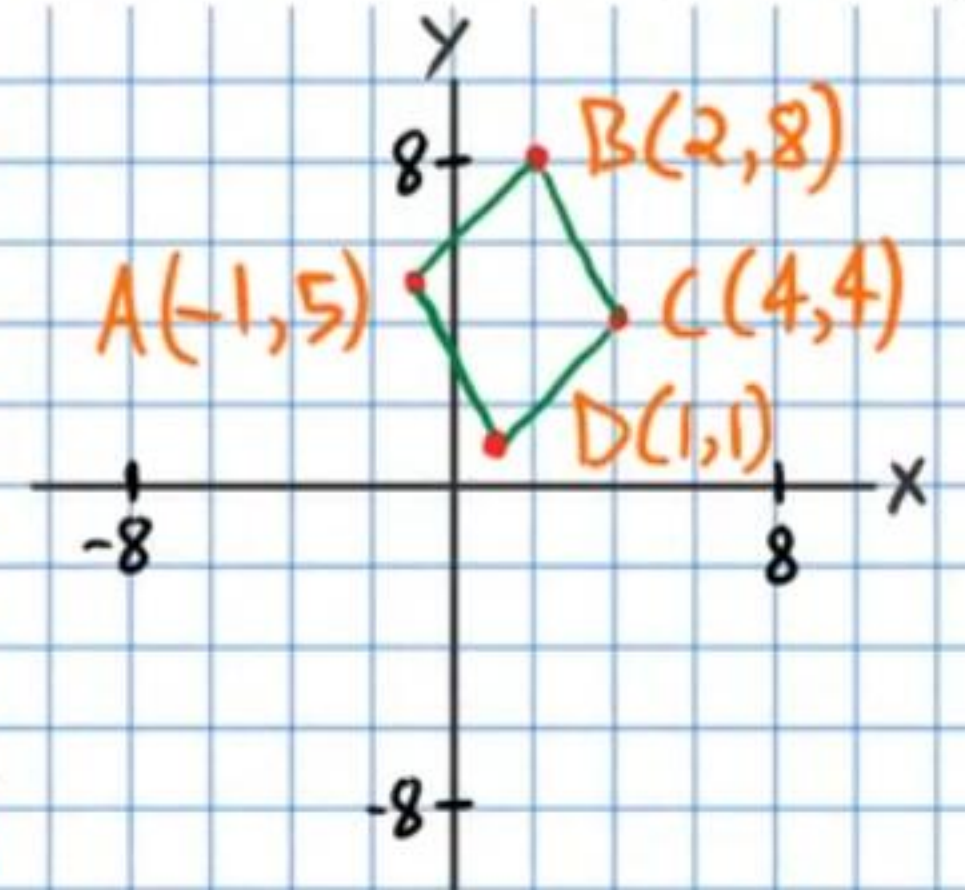
$$m_{AB} = \frac{8 - 5}{2 - (-1)}$$

$$m_{AB} = \frac{3}{3} = 1$$

$$m_{DC} = \frac{4 - 1}{4 - 1} = \frac{3}{3} = 1$$

$$m_{AD} = \frac{1 - 5}{1 - (-1)} = \frac{-4}{2} = -2$$

$$m_{BC} = \frac{4 - 8}{4 - 2} = \frac{-4}{2} = -2$$



Therefore, the quadrilateral is a parallelogram.

⑥ Show that the quadrilateral below is a parallelogram.

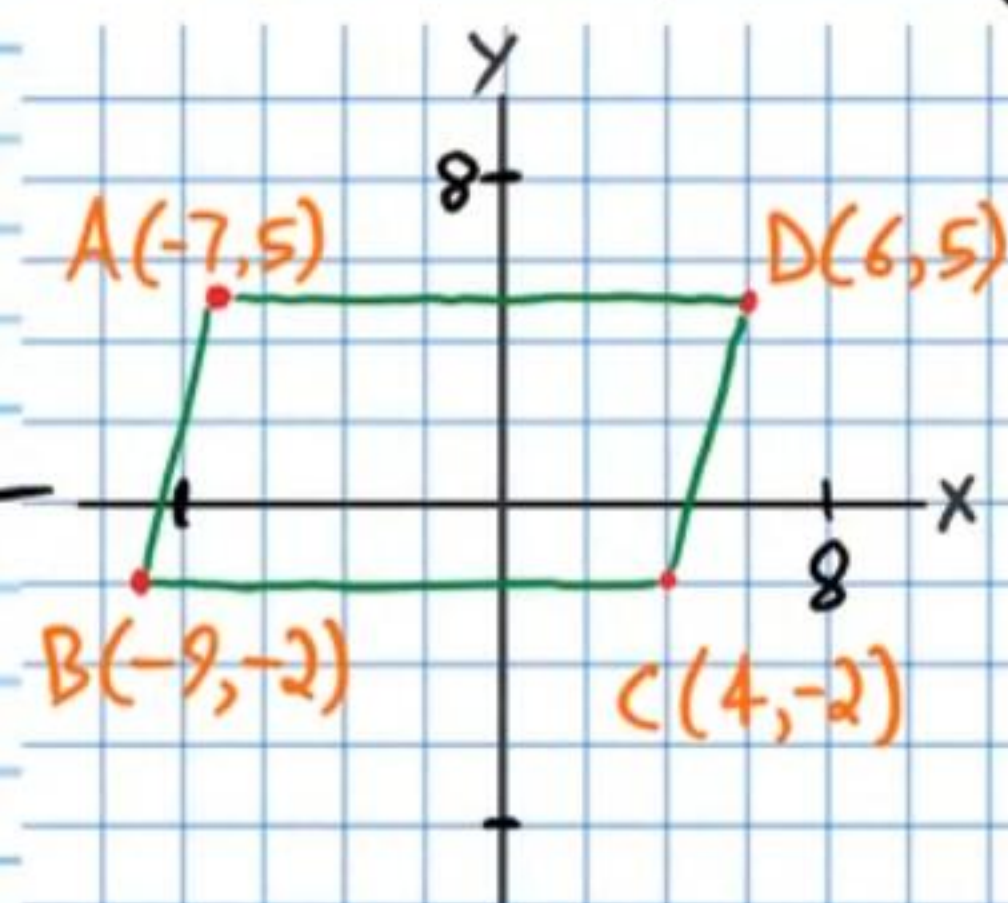
$$M_{AB} = \frac{-2-5}{-9-(-7)} = \frac{-7}{-2} = \frac{7}{2}$$

$$M_{DC} = \frac{-2-5}{4-6} = \frac{-7}{-2} = \frac{7}{2}$$

$$M_{AD} = \frac{5-5}{6-(-7)} = \frac{0}{13} = 0$$

$$M_{BC} = \frac{-2-(-2)}{4-(-9)} = \frac{0}{13} = 0$$

Therefore, the quadrilateral is a parallelogram.



Square: A quadrilateral with all sides congruent and all angle measures 90° .

⑦ Show that the quadrilateral below is a square.

$$D_{AB} = \sqrt{(3-(-3))^2 + (5-3)^2}$$

$$D_{AB} = \sqrt{36 + 4}$$

$$D_{AB} = \sqrt{40} = 2\sqrt{10}$$

$$D_{BE} = \sqrt{(5-3)^2 + (-1-5)^2}$$

$$D_{BE} = \sqrt{4 + 36}$$

$$D_{BE} = \sqrt{40} = 2\sqrt{10}$$

$$D_{AC} = \sqrt{(-1-(-3))^2 + (-3-3)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$$

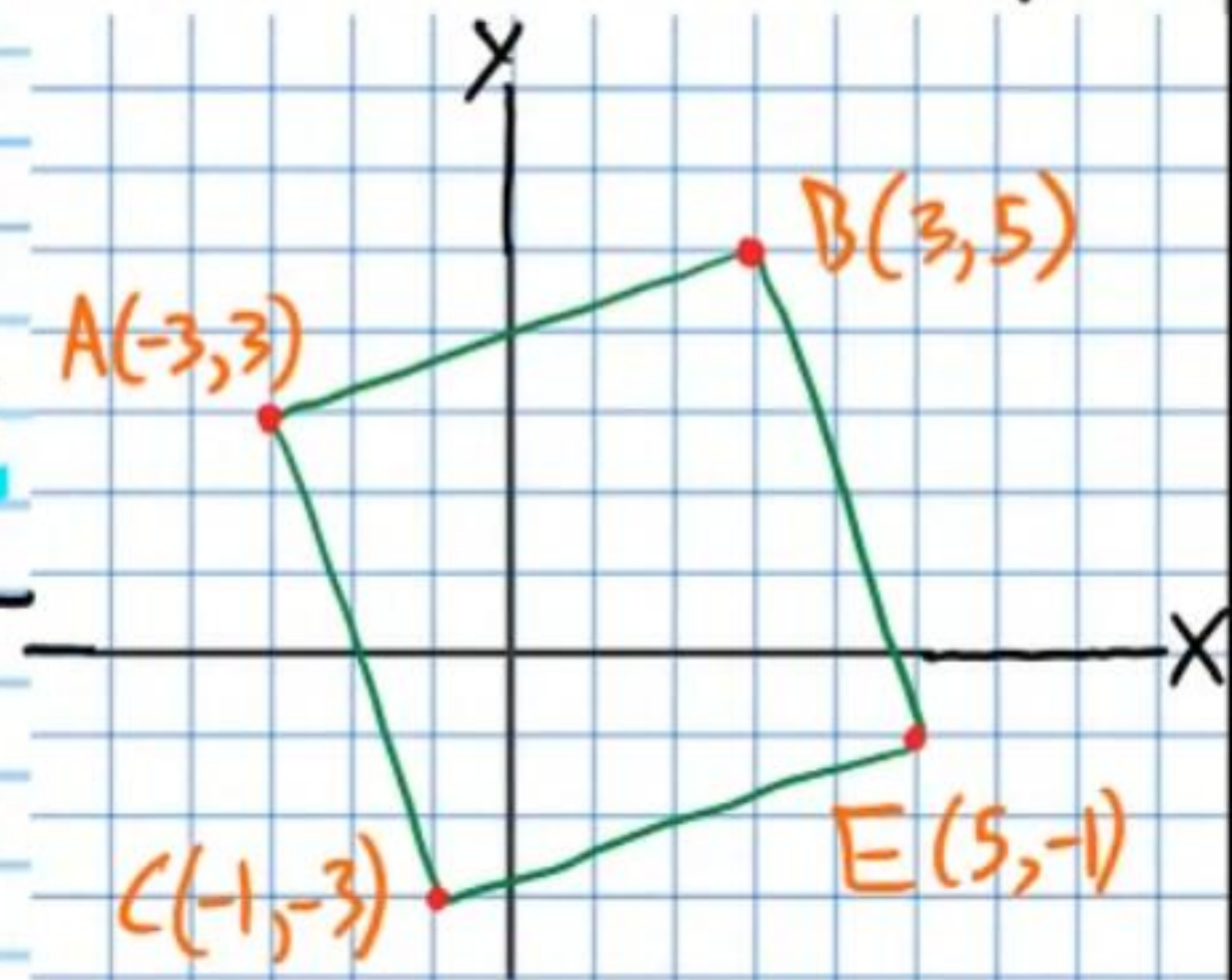
$$D_{CE} = \sqrt{(5-(-1))^2 + (-1-(-3))^2} = \sqrt{36 + 4} = \sqrt{40} = 2\sqrt{10}$$

$$M_{AB} = \frac{5-3}{3-(-3)} = \frac{2}{6} = \frac{1}{3} \quad M_{CE} = \frac{-1-(-3)}{5-(-1)} = \frac{2}{6}$$

$$M_{BE} = \frac{-1-5}{5-3} = \frac{-6}{2} = -3 \quad = \frac{2}{6}$$

$$M_{AC} = \frac{-3-3}{-1-(-3)} = \frac{-6}{2} = -3 \quad = \frac{1}{3}$$

Therefore, the quadrilateral is a square.



⑧ Show that the quadrilateral below is a square.

$$DEG = \sqrt{(3-2)^2 + (5-2)^2}$$

$$DEG = \sqrt{1+9} = \sqrt{10}$$

$$D_{GH} = \sqrt{(6-3)^2 + (4-5)^2}$$

$$D_{GH} = \sqrt{9+1} = \sqrt{10}$$

$$D_{HF} = \sqrt{(5-6)^2 + (1-4)^2}$$

$$D_{HF} = \sqrt{1+9} = \sqrt{10}$$

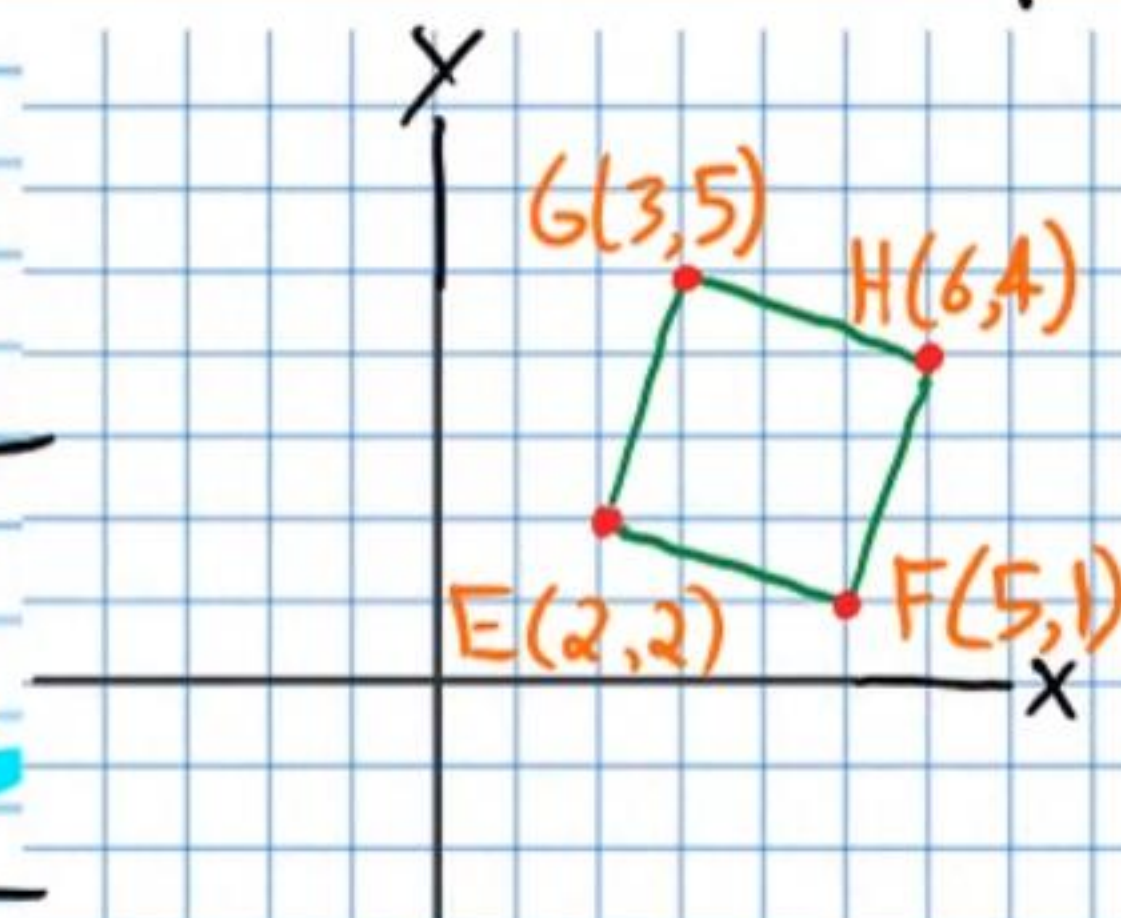
$$D_{EF} = \sqrt{(5-2)^2 + (1-2)^2} = \sqrt{9+1} = \sqrt{10}$$

$$M_{EG} = \frac{5-2}{3-2} = \frac{3}{1} = 3$$

$$M_{GH} = \frac{4-5}{6-3} = \frac{-1}{3}$$

$$M_{EF} = \frac{1-2}{5-2} = \frac{-1}{3}$$

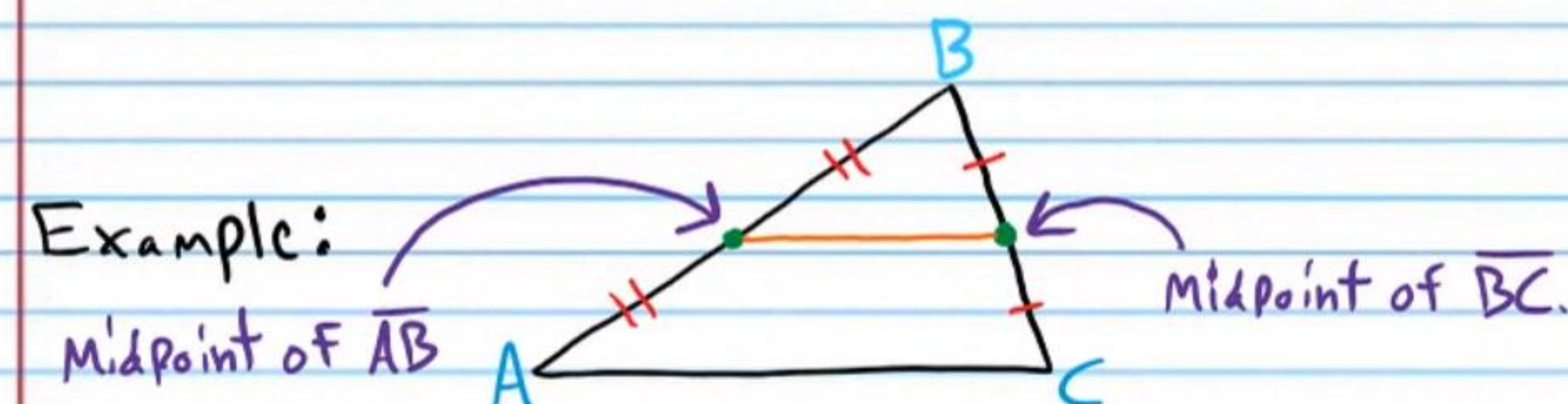
$$M_{FH} = \frac{4-1}{6-5} = \frac{3}{1} = 3$$



Therefore, the quadrilateral is a square.

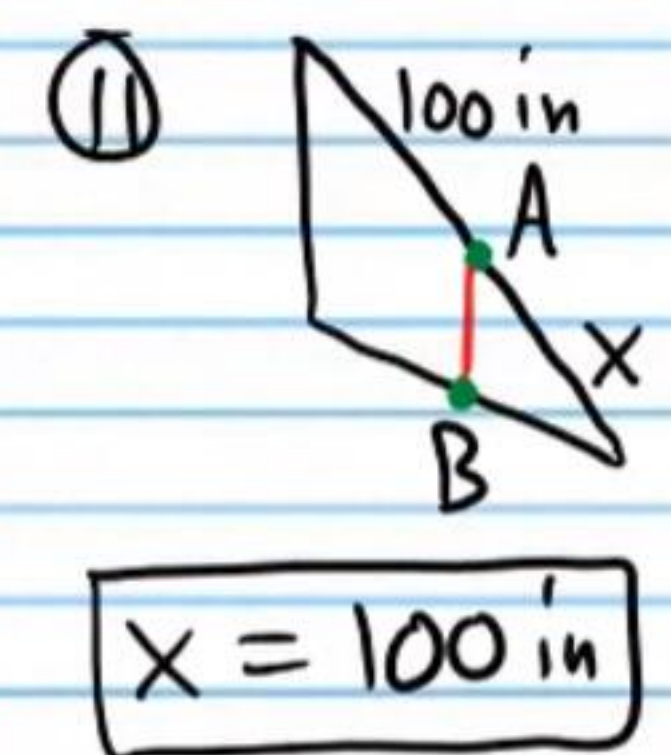
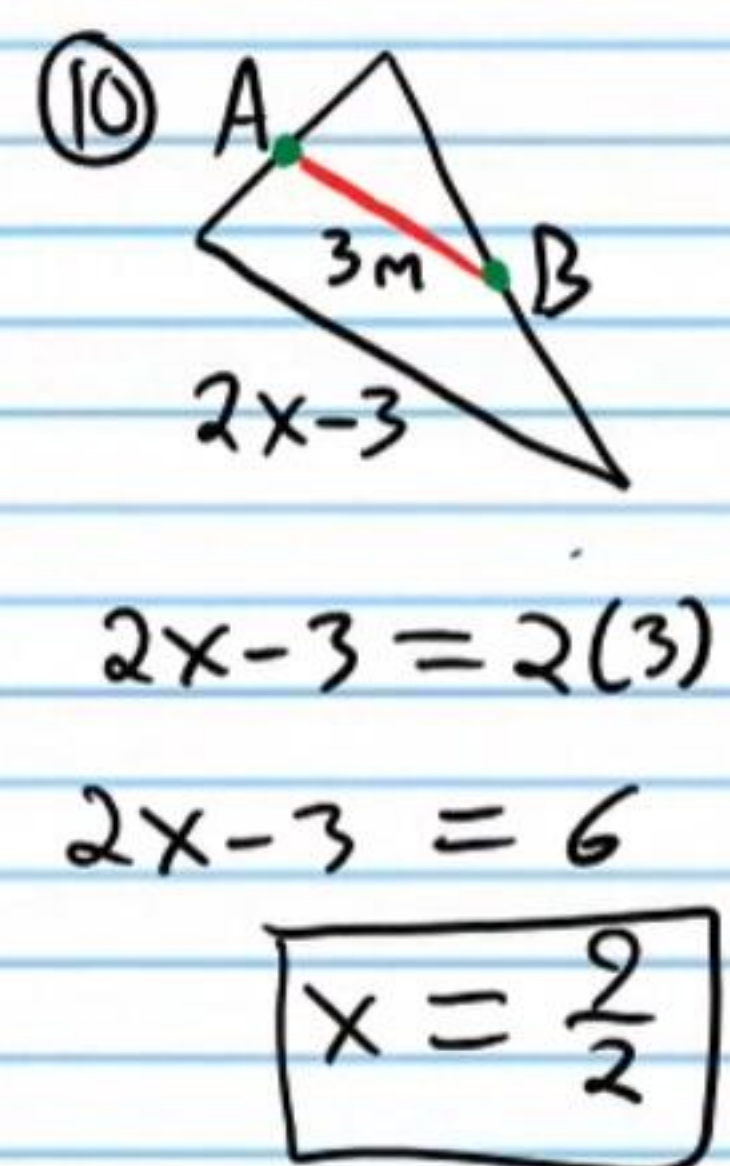
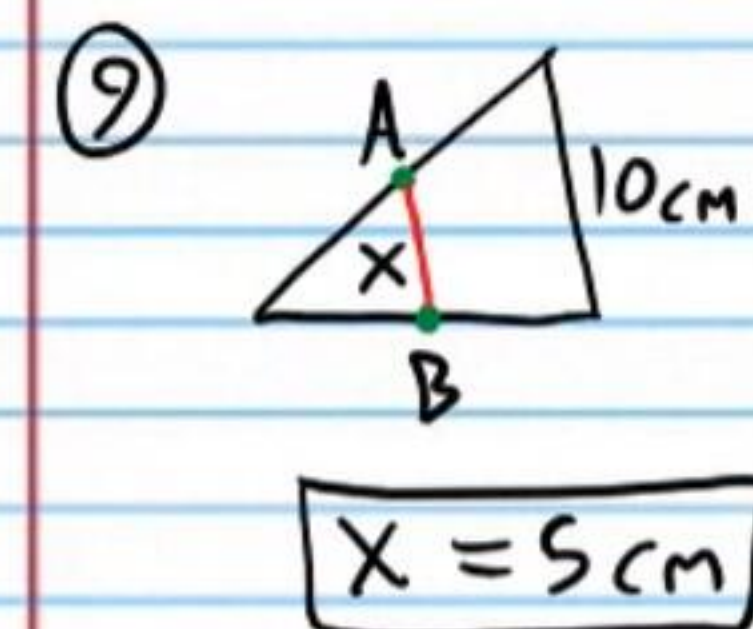
Triangle Midsegment: A segment that connects the midpoints of two sides of a triangle.

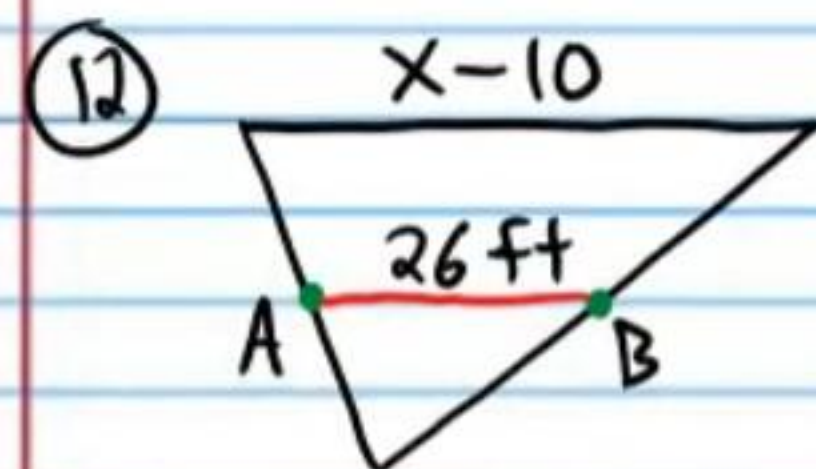
Example:



The Midsegment Theorem: The midsegment of a triangle is half the length of the non-included side and also parallel to the non-included side.

If $\overline{AB}$ is a midsegment of each triangle below, find x .

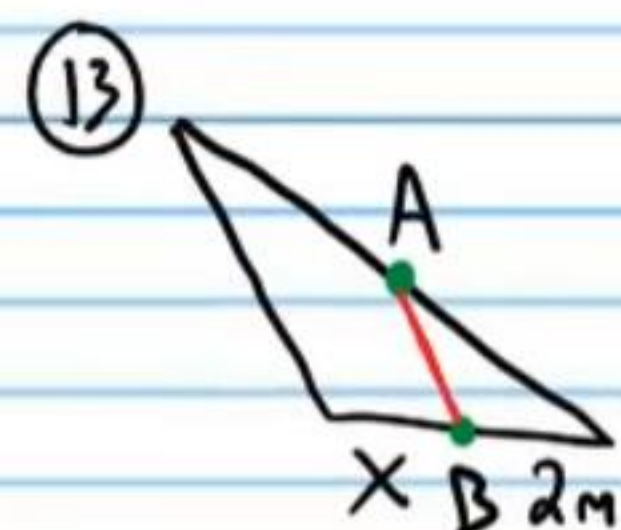




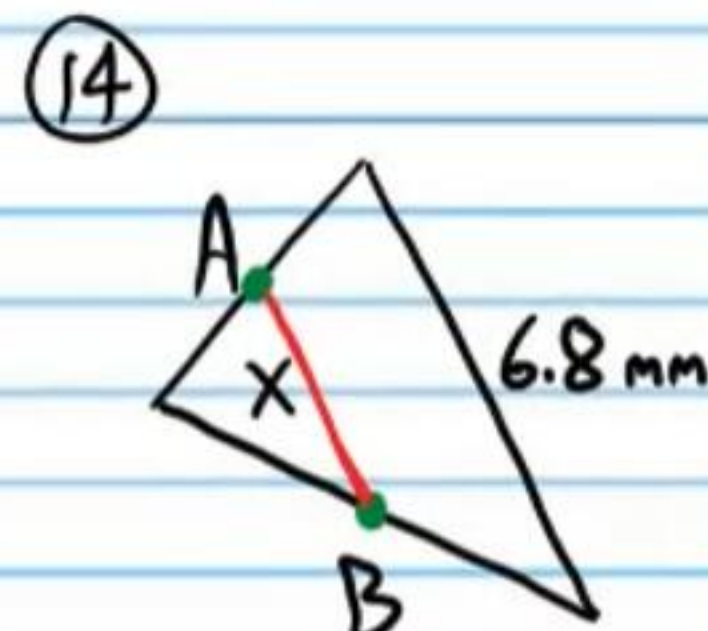
$$X-10 = 2(26)$$

$$X-10 = 52$$

$$\boxed{X = 62}$$



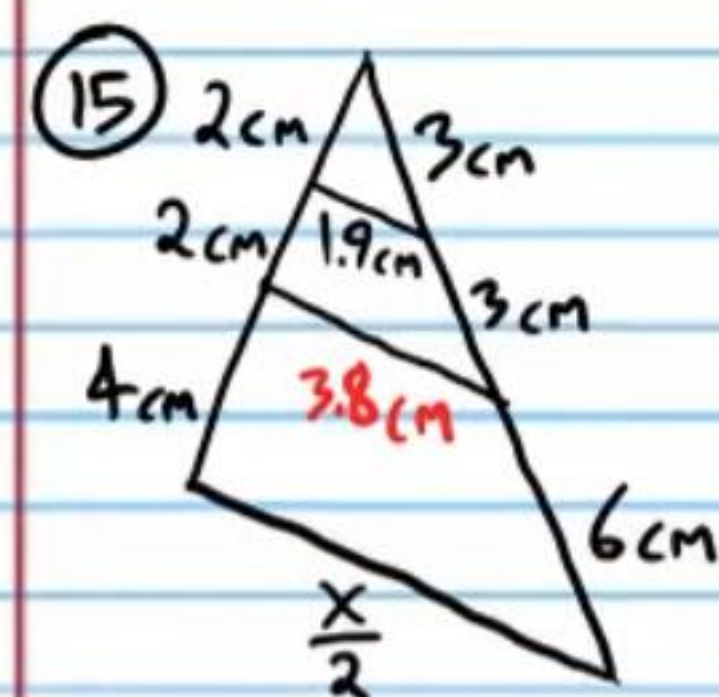
$$\boxed{X = 2m}$$



$$X = \frac{6.8}{2}$$

$$\boxed{X = 3.4 \text{ mm}}$$

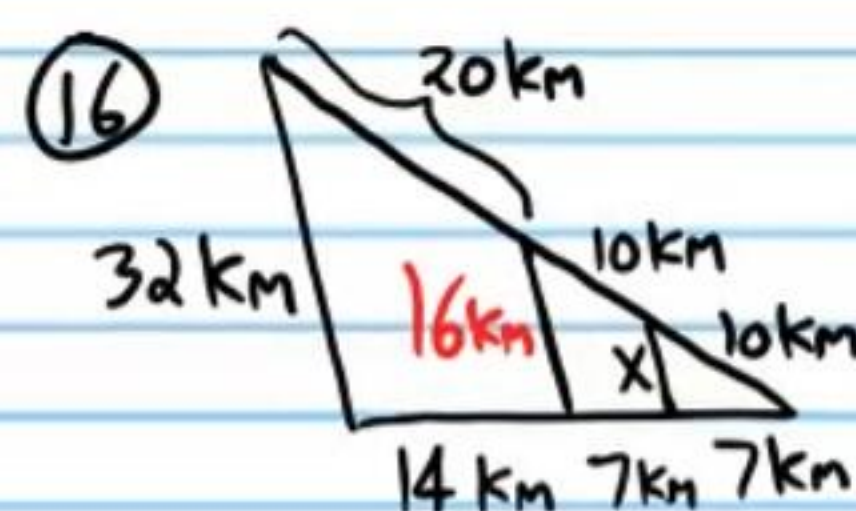
Find X.



$$\frac{X}{2} = 2(3.8)$$

$$\frac{X}{2} = 7.6$$

$$\boxed{X = 15.2}$$



$$\boxed{X = 8 \text{ km}}$$

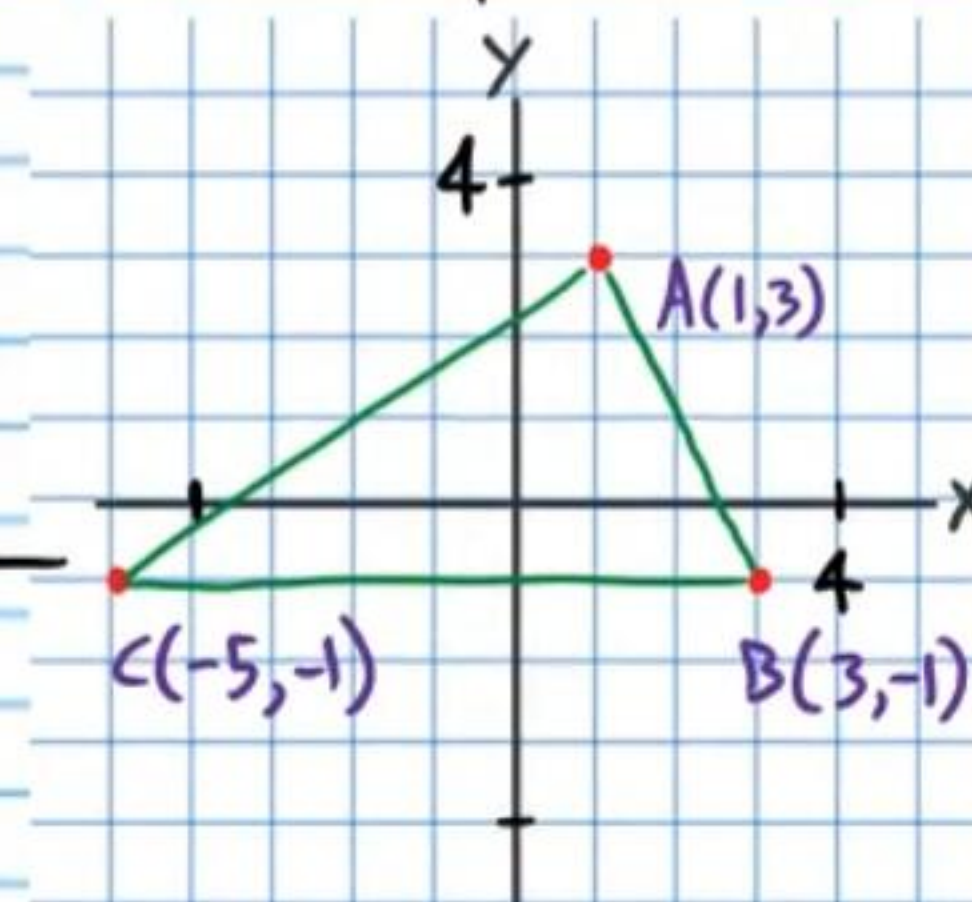
⑰ Show that the midsegment connecting $\overline{CA}$ and $\overline{AB}$ is half the length of $\overline{CB}$ and parallel to $\overline{CB}$ as well.

$$M_{CA} \left(\frac{-5+1}{2}, \frac{-1+3}{2} \right)$$

$$M_{CA}(-2, 1)$$

$$M_{AB} \left(\frac{3+1}{2}, \frac{3+(-1)}{2} \right)$$

$$M_{AB}(2, 1)$$



$$D_M = \sqrt{(2 - (-2))^2 + (1 - 1)^2} = \sqrt{16 + 0} = \sqrt{16} = 4$$

$$D_{CB} = \sqrt{(3 - (-5))^2 + (-1 - (-1))^2} = \sqrt{64 + 0} = \sqrt{64} = 8$$

$$m_{mid} = \frac{1 - 1}{2 - (-2)} = \frac{0}{4} = 0$$

$$m_{CB} = \frac{-1 - (-1)}{3 - (-5)} = \frac{0}{8} = 0$$

Therefore, the Midsegment Theorem holds.

⑮ Show that the midsegment connecting $\overline{CA}$ and $\overline{AB}$ is half the length of $\overline{CB}$ and parallel to $\overline{CB}$ as well.

$$M_{AB} \left(\frac{-3+5}{2}, \frac{2+4}{2} \right)$$

$$M_{AB} (1, 3)$$

$$M_{AC} \left(\frac{-3+(-1)}{2}, \frac{2+0}{2} \right)$$

$$M_{AC} (-2, 1)$$

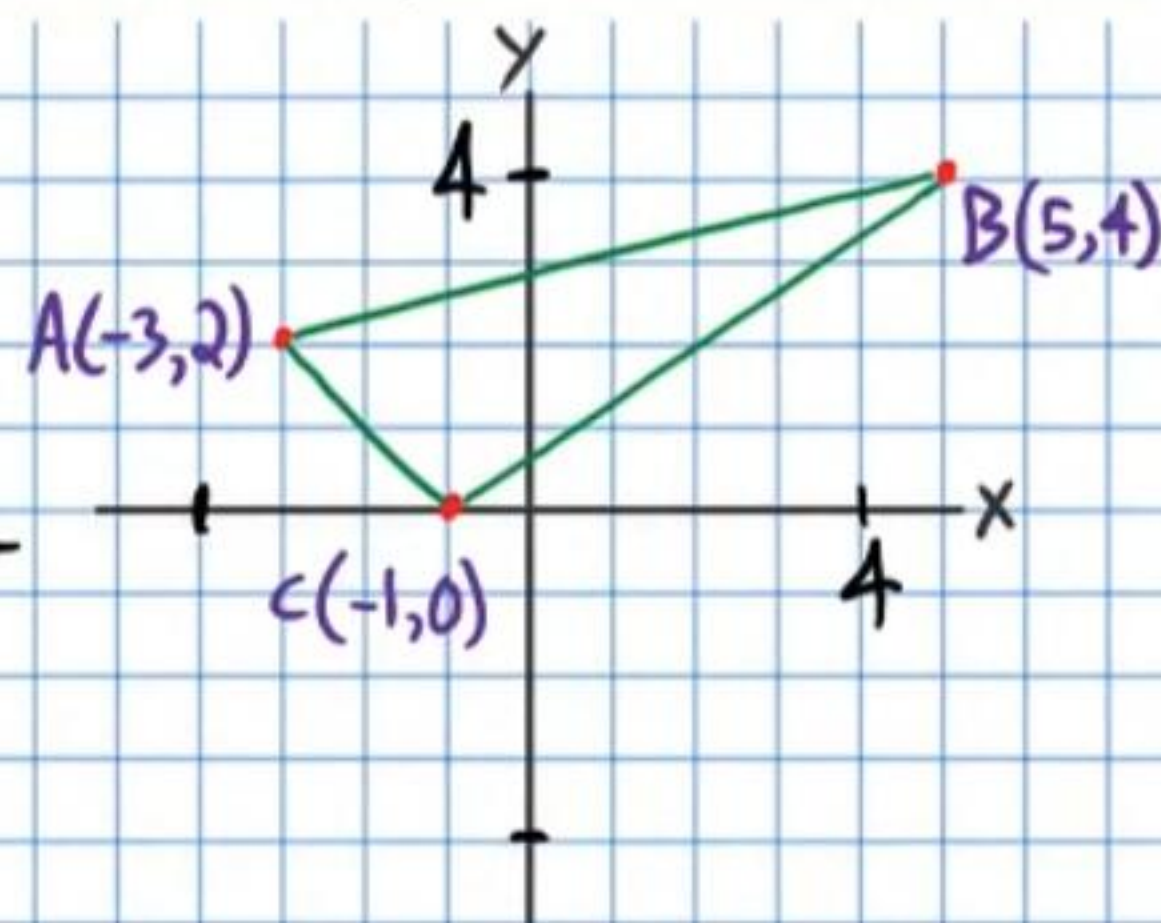
$$D_{mid} = \sqrt{(-2-1)^2 + (1-3)^2} = \sqrt{9+4} = \sqrt{13}$$

$$D_{CB} = \sqrt{(5-(-1))^2 + (4-0)^2} = \sqrt{36+16} = \sqrt{52}$$

$$M_{mid} = \frac{1-3}{-2-1} = \frac{-2}{-3} = \frac{2}{3} \quad \begin{aligned} &= \sqrt{4 \cdot 13} \\ &= \sqrt{4} \cdot \sqrt{13} \end{aligned}$$

$$M_{CB} = \frac{4-0}{5-(-1)} = \frac{4}{6} = \frac{2}{3} \quad = 2\sqrt{13}$$

Therefore, the Midsegment Theorem holds.



⑯ Show that the midsegment connecting $\overline{FE}$ and $\overline{QE}$ is half the length of $\overline{FQ}$ and parallel to $\overline{FQ}$ as well.

$$M_{FE} \left(\frac{-6+8}{2}, \frac{2+(-4)}{2} \right)$$

$$M_{FE} (1, -1)$$

$$M_{QE} \left(\frac{2+8}{2}, \frac{8+(-4)}{2} \right)$$

$$M_{QE} (5, 2)$$

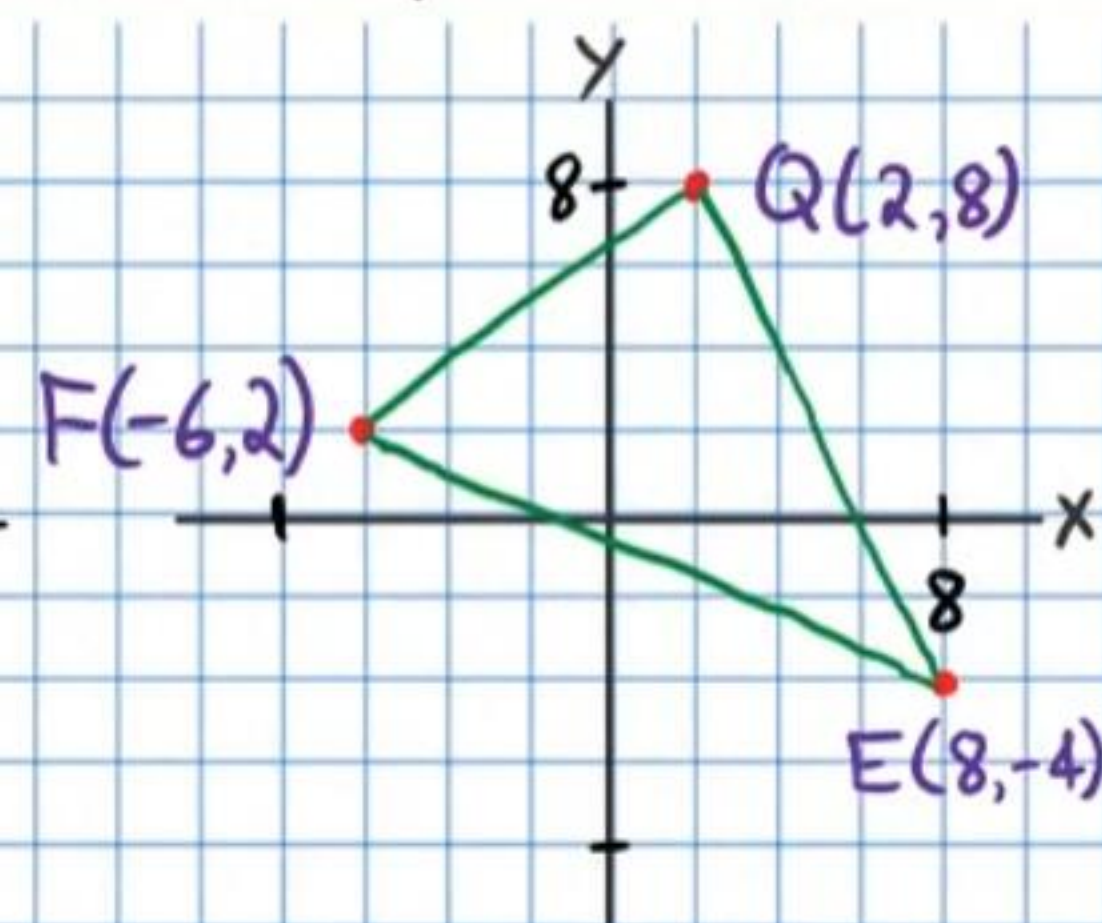
$$D_{mid} = \sqrt{(5-1)^2 + (2-(-1))^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$$D_{FQ} = \sqrt{(2-(-6))^2 + (8-2)^2} = \sqrt{64+36} = \sqrt{100}$$

$$M_{mid} = \frac{2-(-1)}{5-1} = \frac{3}{4} \quad = 10$$

$$M_{FQ} = \frac{8-2}{2-(-6)} = \frac{6}{8} = \frac{3}{4}$$

Therefore, the Midsegment Theorem holds.



② Show that the midsegment connecting $\overline{FQ}$ and $\overline{FE}$ is half the length of $\overline{QE}$ and parallel to $\overline{QE}$ as well.

$$M_{FQ} \left(\frac{4+(-8)}{2}, \frac{4+2}{2} \right)$$

$$M_{FQ} (-2, 3)$$

$$M_{FE} \left(\frac{-8+8}{2}, \frac{2+(-4)}{2} \right)$$

$$M_{FE} (0, -1)$$

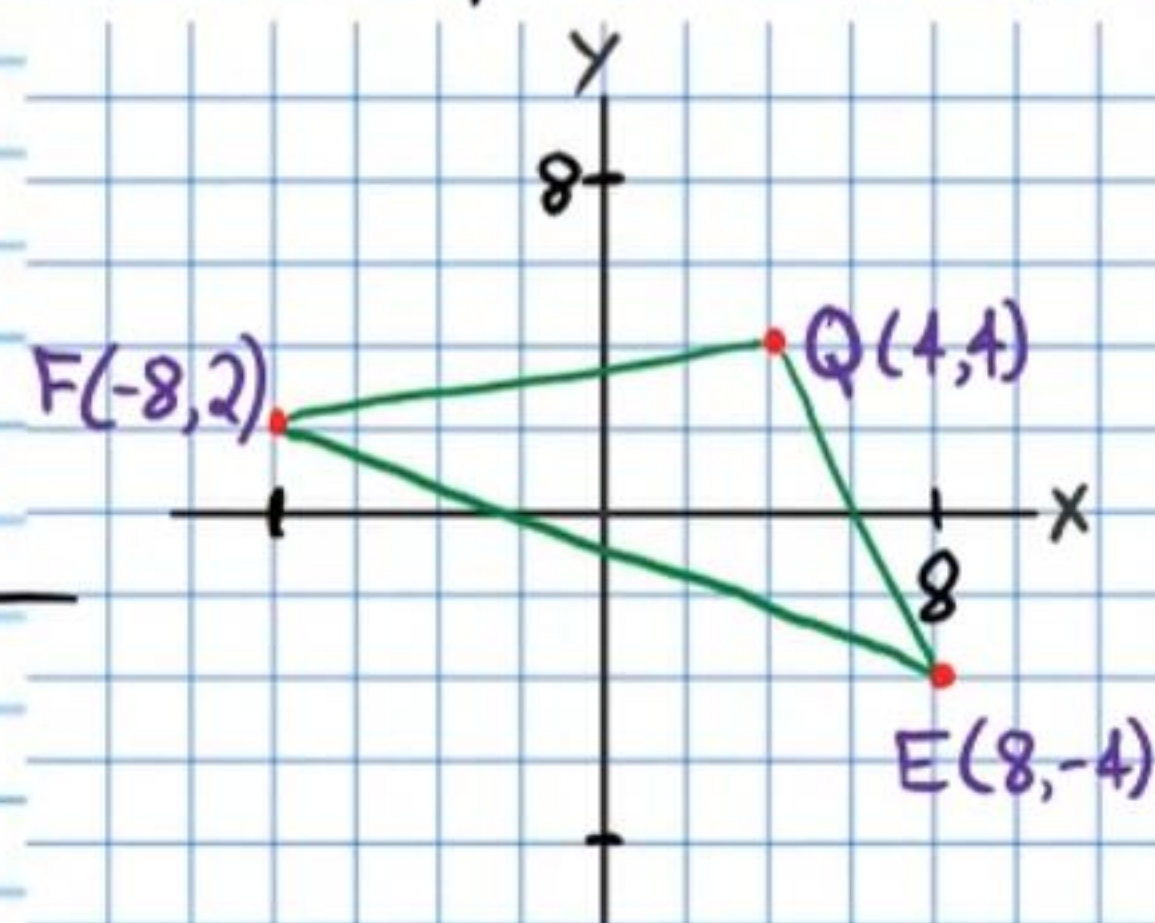
$$D_{mid} = \sqrt{(0-(-2))^2 + (-1-3)^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$$

$$D_{QE} = \sqrt{(8-4)^2 + (-4-4)^2} = \sqrt{16+64} = \sqrt{80} = \sqrt{5 \cdot 16}$$

$$m_{mid} = \frac{-1-3}{0-(-2)} = \frac{-4}{2} = -2$$

$$m_{QE} = \frac{-4-4}{8-4} = \frac{-8}{4} = -2$$

Therefore, the Midsegment Theorem holds.



Coordinate Proofs with Variables

② Write a coordinate proof to show that the Midsegment Theorem is true.

$$F \left(\frac{0+h}{2}, \frac{0+k}{2} \right) = F \left(\frac{h}{2}, \frac{k}{2} \right)$$

$$E \left(\frac{h+m}{2}, \frac{k+0}{2} \right) = E \left(\frac{h+m}{2}, \frac{k}{2} \right)$$

$$D_{FE} = \sqrt{\left(\frac{h+m}{2} - \frac{h}{2} \right)^2 + \left(\frac{k}{2} - \frac{k}{2} \right)^2} = \sqrt{\left(\frac{m}{2} \right)^2 + 0^2}$$

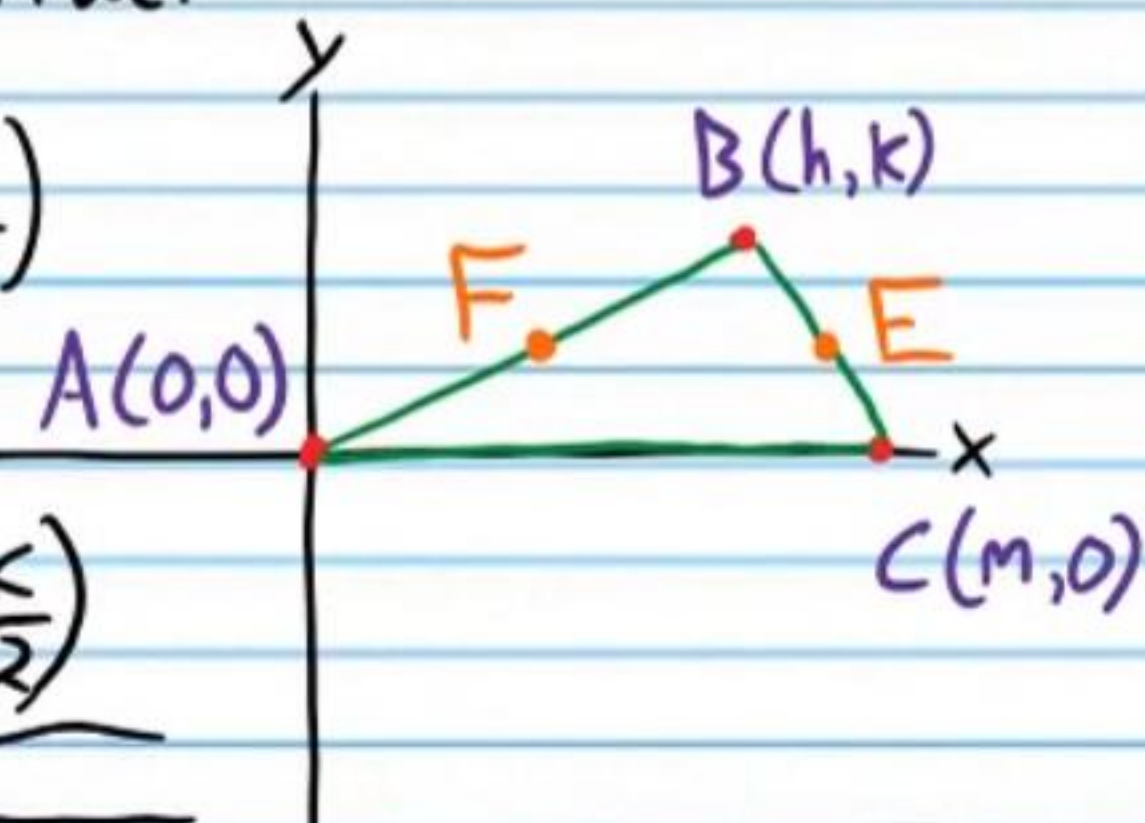
$$D_{AC} = \sqrt{(m-0)^2 + (0-0)^2} = \sqrt{m^2} = m$$

$$D_{AC} = \sqrt{m^2 + 0^2} = \sqrt{m^2} = m$$

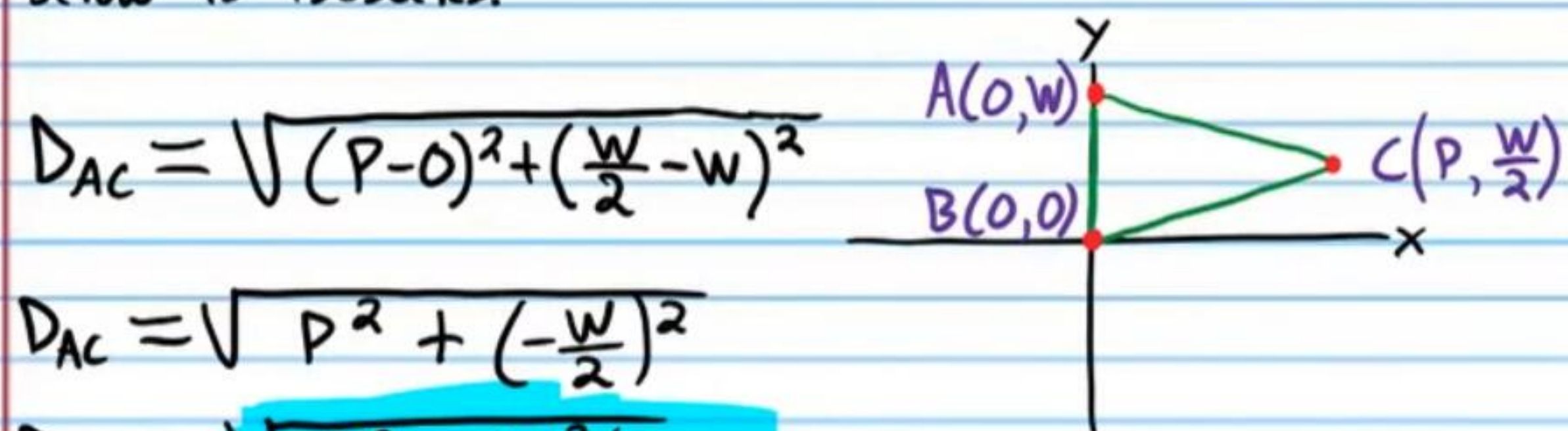
$$m_{FE} = \frac{\frac{k}{2} - \frac{k}{2}}{\frac{h+m}{2} - \frac{h}{2}} = 0$$

$$m_{AC} = \frac{0-0}{m-0} = \frac{0}{m} = 0$$

Therefore, the Midsegment Theorem is true.



② Write a coordinate proof to show that the triangle below is isosceles.



$$D_{AC} = \sqrt{(p-0)^2 + (\frac{w}{2}-w)^2}$$

$$D_{AC} = \sqrt{p^2 + (-\frac{w}{2})^2}$$

$$D_{AC} = \sqrt{p^2 + w^2/4}$$

$$D_{BC} = \sqrt{(p-0)^2 + (\frac{w}{2}-0)^2} = \sqrt{p^2 + (\frac{w}{2})^2}$$

$$= \sqrt{p^2 + w^2/4}$$

Therefore, $\triangle ABC$ is isosceles.

③ Write a coordinate proof to show that the quadrilateral below is a parallelogram (i.e. two pairs of parallel sides).

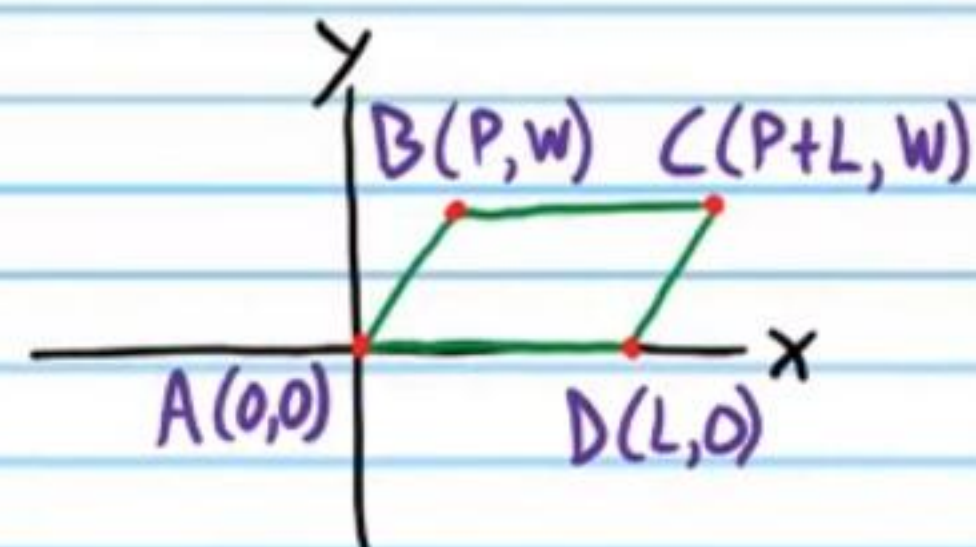
$$M = \frac{y_2 - y_1}{x_2 - x_1}$$

$$M_{AB} = \frac{w-0}{p-0} = \frac{w}{p}$$

$$M_{DC} = \frac{w-0}{p+l-l} = \frac{w}{p}$$

$$M_{BC} = \frac{w-w}{p+l-p} = \frac{0}{l} = 0$$

$$M_{AD} = \frac{0-0}{l-0} = \frac{0}{l} = 0$$

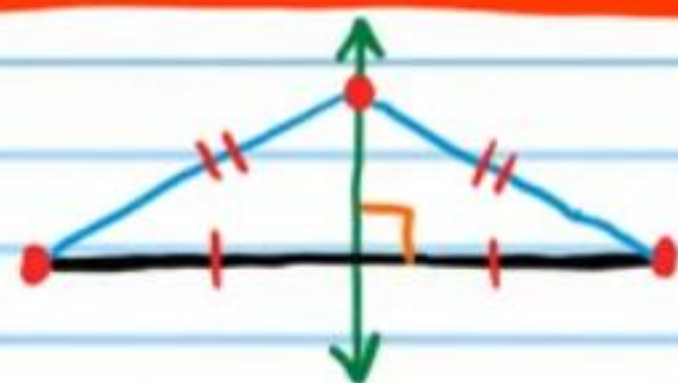


Therefore, the quadrilateral is a parallelogram.

Bisector Theorems

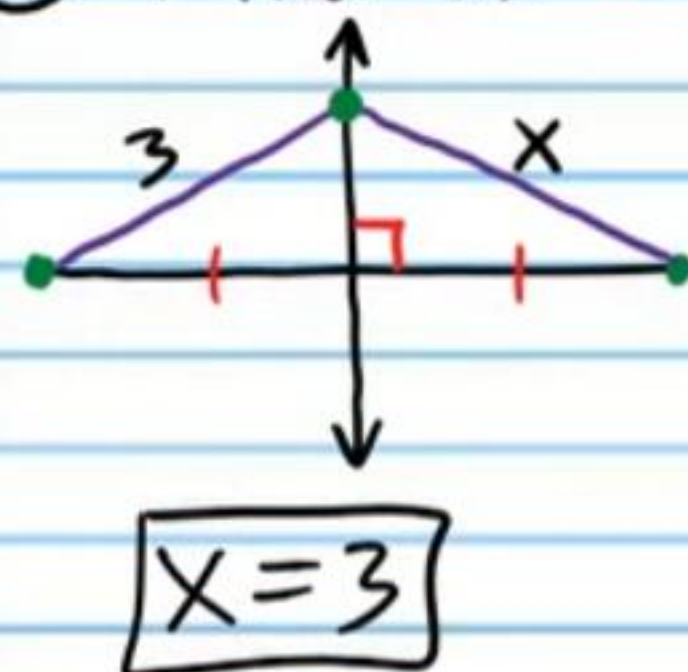
The Perpendicular Bisector Theorem: If a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment.

Example:



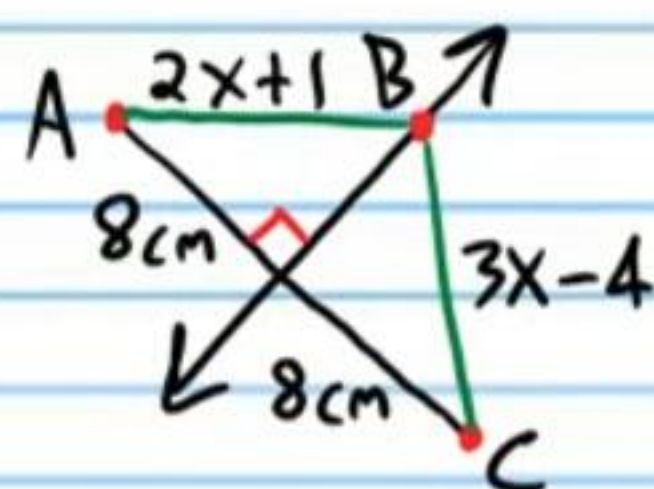
The Perpendicular Bisector Converse Theorem: If a point is equidistant from the endpoints of a segment, then it is on the perpendicular bisector of the segment.

②④ Find X.



$$X = 3$$

②⑤ Find BC.



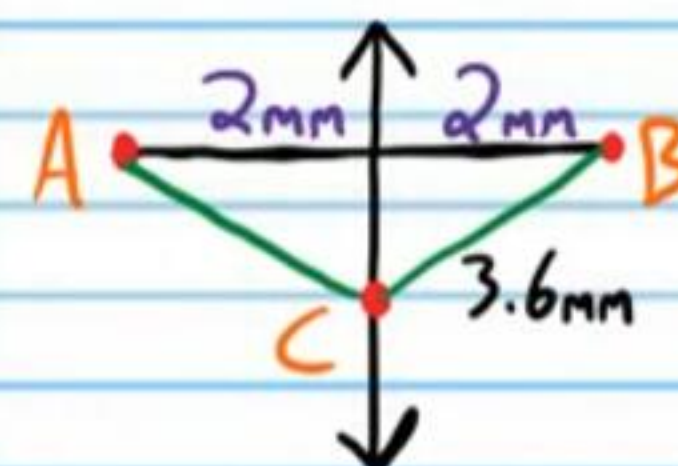
$$2x+1 = 3x-4$$

$$1 = x-4$$

$$5 = x$$

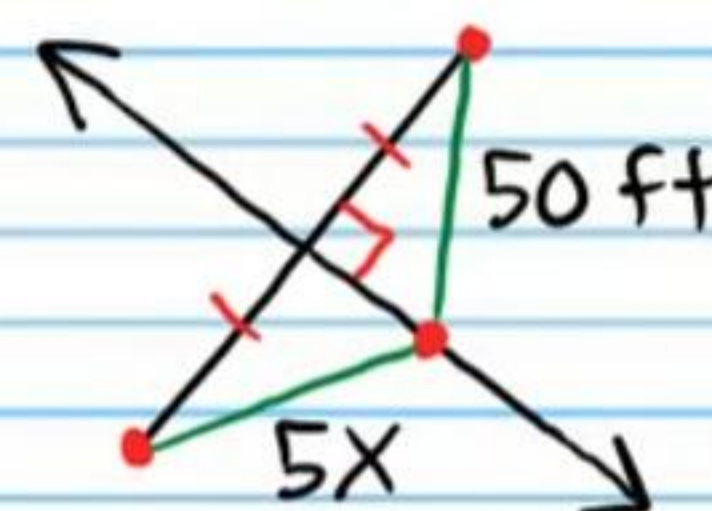
$$BC = 3(5) - 4 = 11$$

②⑥ Find AC.



Not enough information.

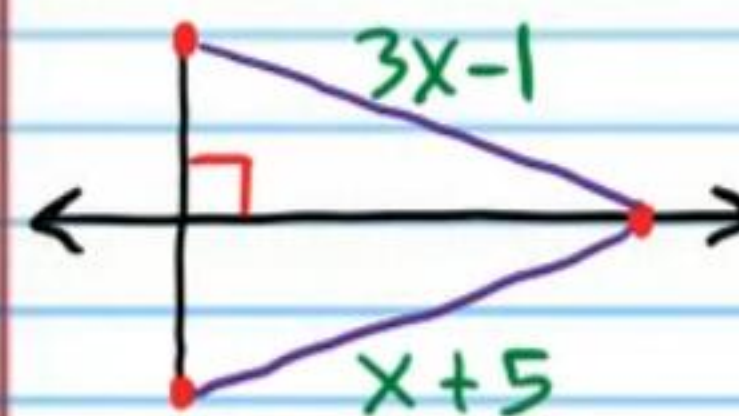
②⑦ Find X.



$$50 = 5X$$

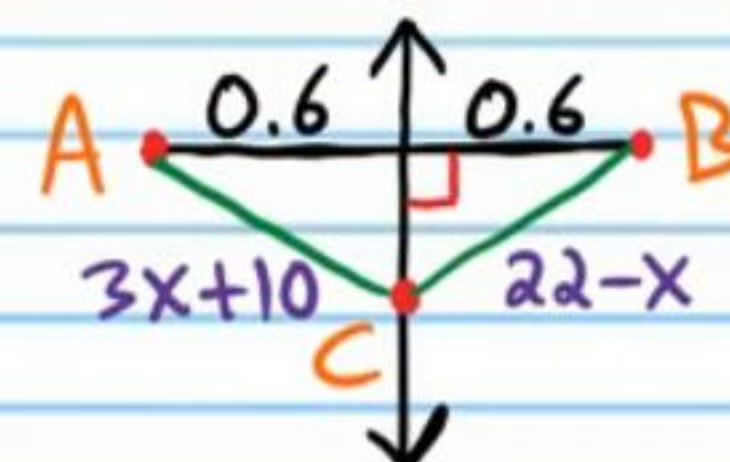
$$10 = X$$

②⑧ Find X.



Not enough information.

②⑨ Find AC.



$$3x+10 = 22-x$$

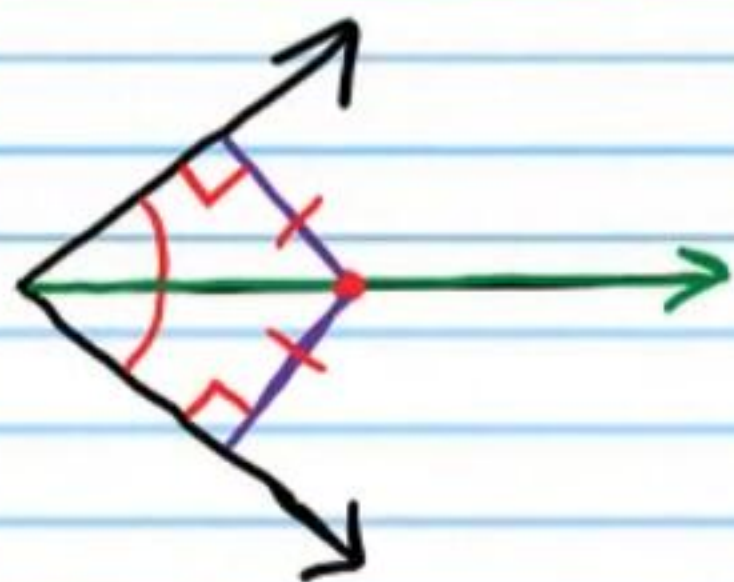
$$4x+10 = 22$$

$$4x = 12$$

$$x = 3$$

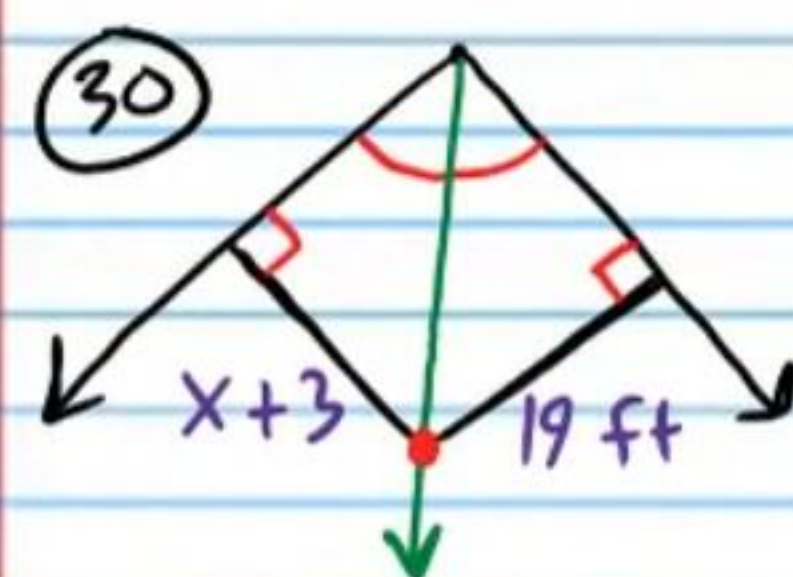
$$AC = 3(3) + 10 = 19$$

The Angle Bisector Theorem: If a point is on the bisector of an angle, then it is equidistant from the rays of the angle.



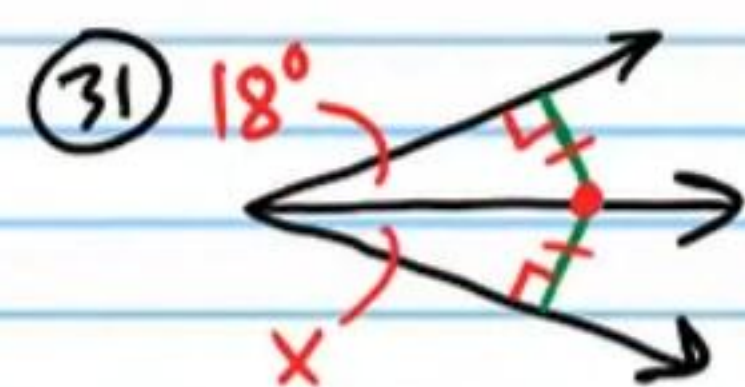
The Angle Bisector Converse Theorem: If a point is on the interior of an angle and it is equidistant from the rays of the angle, then it is on the bisector of the angle.

Find x if possible.

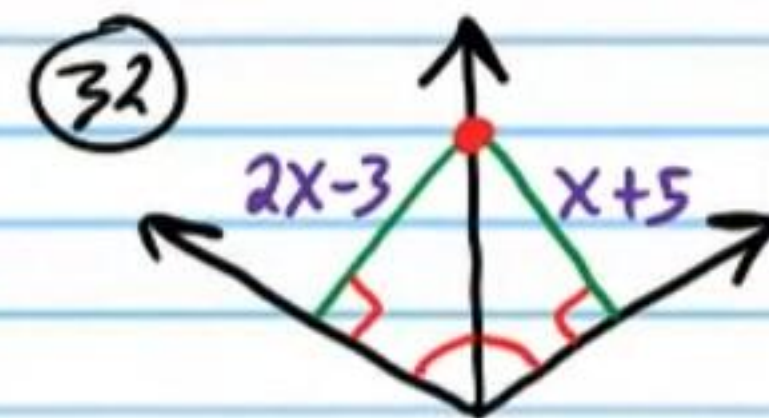


$$x+3=19$$

$$\boxed{x=16}$$



$$\boxed{x=18}$$



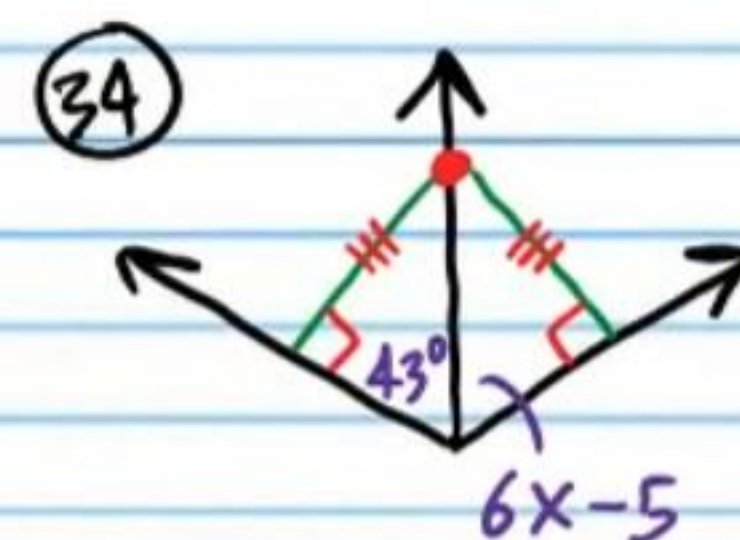
$$2x-3=x+5$$

$$x-3=5$$

$$\boxed{x=8}$$



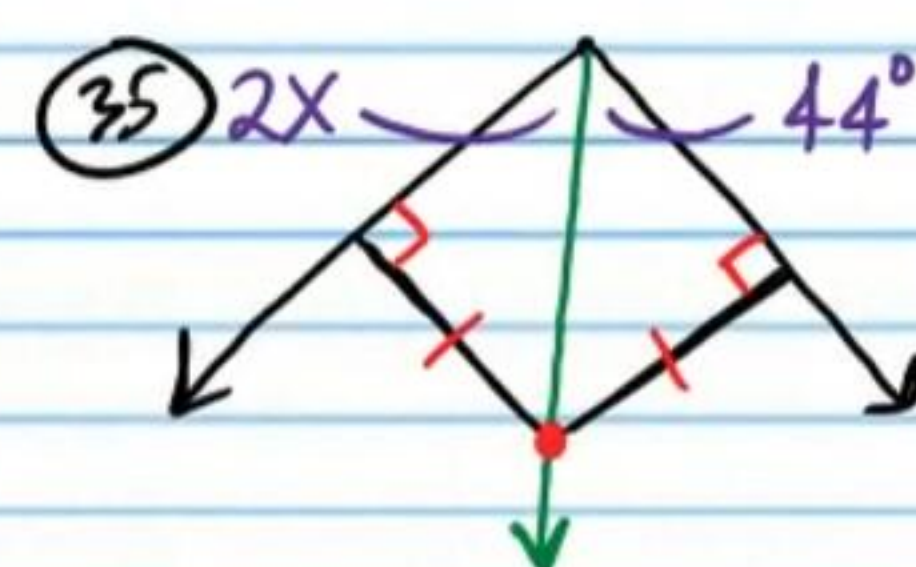
Not enough information



$$43=6x-5$$

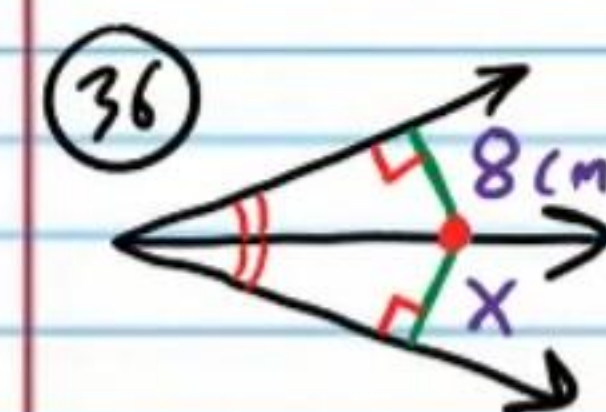
$$48=6x$$

$$\boxed{8=x}$$

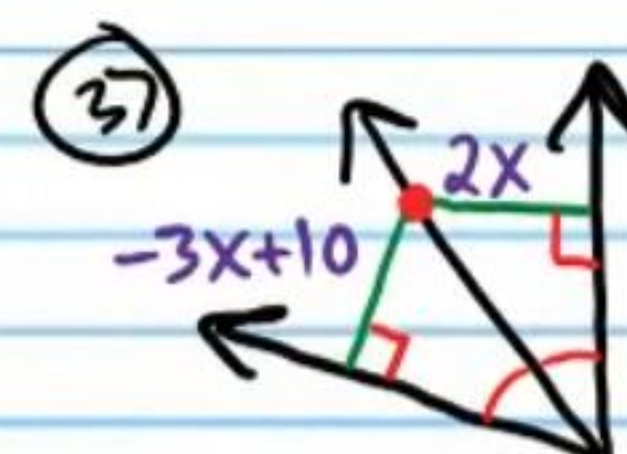


$$2x=44$$

$$\boxed{x=22}$$



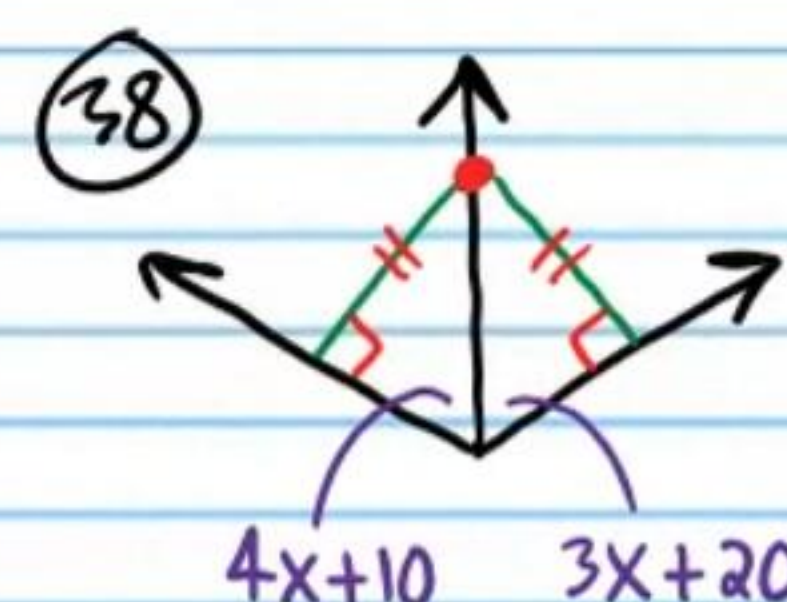
$$\boxed{x=8\text{ cm}}$$



$$-3x+10=2x$$

$$10=5x$$

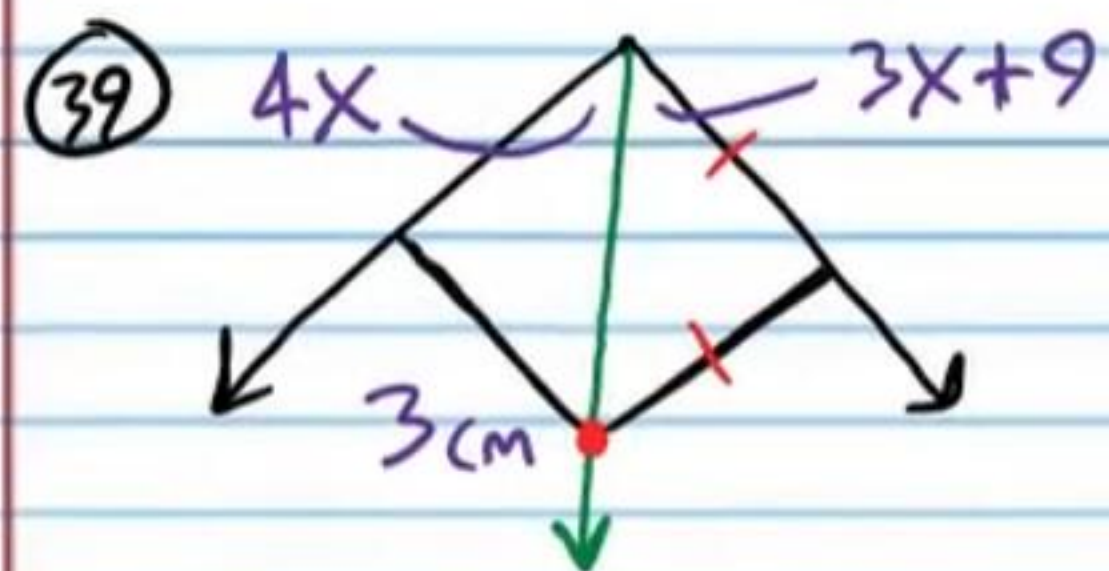
$$\boxed{2=x}$$



$$4x+10=3x+20$$

$$x+10=20$$

$$\boxed{x=10}$$



Not enough information.